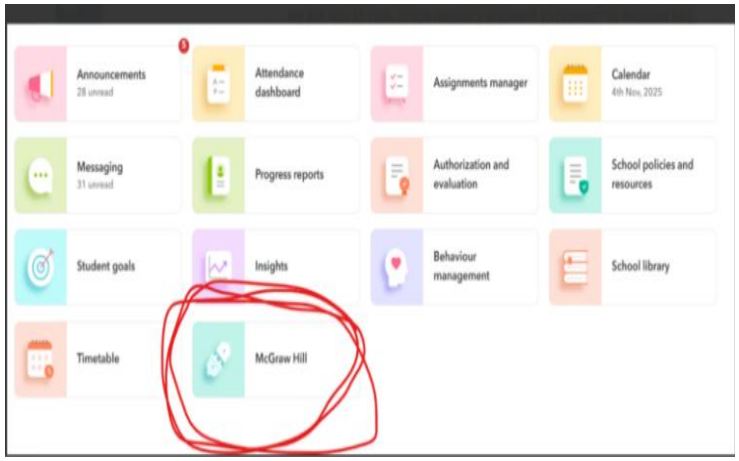
Week 8: 19th - 23rd October 2026

Subject: Mathematics Course 2 | Unit: Ratios, Rates and Proportional Relationships | Grade: 7

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
L1: Represent Proportional Relationships with Equations	I can write an equation of the form y = kx for a proportional relationship and explain what the constant of proportionality means.	8.2 Vary challenge levels with guided steps, differentiated practice, and extension tasks. Formative checks: questioning, copybook/workbook practice, partner explanations, mini whiteboards, and exit tickets. Checks for understanding: Formative assessment covering current and previously taught concepts to monitor students' ongoing progress and understanding.	**Practice resources:** **Lesson 1:** **Represent Proportional Relationships with Equations.** **Lesson 2:** **Describe Proportional Relationships.** **Lesson 3:** **Multi-Step Ratio Problems.** **McGraw Hill through Toddle; slides, worksheets, and homework through Teams.**
L2: Describe Proportional Relationships	I can connect a table, graph, equation, and written description that represent the same proportional relationship.		
L3: Use Proportional Reasoning to Solve Multi-Step Ratio Problems	I can solve multi-step ratio and proportion problems and explain each step using correct units.		
L4: Proportional Relationships Work Period + Checkpoint	I can apply proportional reasoning to my Chill Zone design and clearly communicate my table, graph or equation, constant of proportionality, and calculations.		
Teacher's notes / Reminders	- Make use of MAGMA – as this gives students extra practice. - Teaching and learning resources will be uploaded on teams in the shared files folder. - Please remember to bring your Reveal Math Book – Volume 1 and copybook to class. - McGraw Hill can be accessed through Toddle.		

- For L4: do not forget to bring your Chill Zone GRASPS Task work. Keep all your proportional-reasoning evidence together and organized for the checkpoint.

Mc Graw hill: Can be accessed on Toddle:



- **MAGMA login: Students should know their Microsoft username and password (download the application to sign in with your QR code). Please go to your teacher for assistance.**
  Please make use of MAGMA for extra practice.

Week 8

Technology Integration دمج التكنولوجيا	MAGMA Teams Class notebook/workbook.
Cross-curricular links لروابط بين المواد الدراسية	Science: Speed, density, and other rates; unit conversions in experiments Economics/Business: Discounts, profit, tax, and financial literacy Geography: Scale maps and population density Food Technology: Recipe scaling and proportions
External Assessment Integration دمج التقييمات الخارجية	MAP / Common Core alignment: interpret proportional relationships across tables, graphs, equations, and real-world multi-step contexts. Reinforce reasoning, units, and interpretation of k.
Wellbeing/ ATL Skills النفسية/ أساليب التعلم	Thinking: connect multiple representations and select efficient strategies. Communication: explain k, equations, and multi-step reasoning using correct mathematical language. Self-management: organize calculations and checkpoint evidence; check units and reasonableness. Social: compare representations, give peer feedback, and improve clarity.
Standards المعايير	7.RP.A.2; 7.RP.A.2.b; 7.RP.A.2.c; 7.RP.A.2.d; 7.RP.A.3
Inquiry/ TOK Question البحث والاستقصاء أسئلة	L1: How can an equation represent a proportional relationship? L2: Why do mathematicians and designers use more than one representation? L3: How can proportional reasoning help us make fair and accurate planning decisions? L4: How can we communicate proportional reasoning so another person can understand and use our design information?
Success Criteria (Measurable) معايير النجاح	Lesson 1: ■ All students must identify k and write an equation y = kx from a simple proportional relationship. ● Most students should use the equation to find an unknown value and explain what k means in context. ▲ Some students could create an equation from a real UAE planning context and justify how the equation represents the relationship. Lesson 2: ■ All students must match a table, graph, equation, and written description for the same proportional relationship. ● Most students should translate between at least three representations and identify the same k in each. ▲ Some students could explain which representation communicates the relationship most effectively for a chosen audience and why. Lesson 3: ■ All students must solve a multi-step proportional problem and include correct units. ● Most students should explain the sequence of steps and use proportional reasoning to find all required quantities.

	▲ Some students could compare two solution strategies, check reasonableness, and justify the more efficient method. Lesson 4: ■ All students must present one proportional table and one matching graph or equation. ● Most students should explain the constant of proportionality and complete one accurate multi-step calculation for the Chill Zone. ▲ Some students could evaluate whether their representation and calculations are realistic, revise them, and justify their improvements.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Respect الاحترام Responsibility المسؤولية Sustainability الاستدامة
	Area: UAE innovation, sustainability, and community-space design. Students use proportional reasoning to make realistic المجال .and responsible decisions about cost, materials, water, seating, pathways, and shared spaces
	Implementation: Students write and interpret equations for UAE planning contexts, compare tables, graphs, equations, and التطبيق descriptions, solve multi-step resource-planning problems, and apply the mathematics directly to their Chill Zone proposal and .checkpoint evidence